

Evaluating Triple Iterated Integrals in Exercise #7-20.

07) $\int_0^1 \int_0^1 \int_0^1 (x^2 + y^2 + z^2) dz dy dx$

Solution:

$$\int_0^1 \int_0^1 \left[\int_0^1 (x^2 + y^2 + z^2) dz \right] dy dx$$

$$\int_0^1 \int_0^1 \left[x^2 \int_0^1 dz + y^2 \int_0^1 dz + \int_0^1 z^2 dz \right] dy dx$$

$$\int_0^1 \int_0^1 \left[x^2 \cdot z + y^2 \cdot z + \frac{z^3}{3} \right]_0^1 dy dx$$

$$\int_0^1 \int_0^1 \left(x^2(1) + y^2(1) + \frac{1}{3} \right) dy dx$$

$$\int_0^1 \left[x^2 \int_0^1 dy + \int_0^1 y^2 dy + \frac{1}{3} \int_0^1 dy \right] dx$$

$$\int_0^1 \left[x^2 y + \frac{y^3}{3} + \frac{1 \cdot y}{3} \right]_0^1 dx$$

$$\int_0^1 \left(x^2(1) + \frac{(1)^3}{3} + \frac{1 \cdot (1)}{3} \right) dx$$

$$\int_0^1 \left(x^2 + \frac{2}{3} \right) dx$$

$$\int_0^1 x^2 + \frac{2}{3} \int_0^1 dx$$

$$\left[\frac{x^3}{3} + \frac{2 \cdot x}{3} \right]_0^1$$

$$\frac{(1)^3}{3} + \frac{2(1)}{3}$$

$$\frac{1}{3} + \frac{2}{3}$$

$$\frac{1+2}{3} = \frac{3}{3} = 1 \quad \text{Answer}$$

08) $\int_0^{\sqrt{2}} \int_0^{3y} \int_{x^2+3y^2}^{8-x^2-y^2} dz dndy$

Solution:

$$\int_0^{\sqrt{2}} \int_0^{3y} \left[\int_{x^2+3y^2}^{8-x^2-y^2} dz \right] dndy$$

$$\int_0^{\sqrt{2}} \int_0^{3y} \left[z \right]_{x^2+3y^2}^{8-x^2-y^2} dndy$$

$$\int_0^{\sqrt{2}} \int_0^{3y} \left[(8-x^2-y^2) - (x^2+3y^2) \right] dndy$$

$$\int_0^{\sqrt{2}} \int_0^{3y} (8-x^2-y^2-x^2-3y^2) dndy$$

$$\int_0^{\sqrt{2}} \int_0^{3y} (8-2x^2-4y^2) dndy$$

$$\int_0^{\sqrt{2}} \left[8 \int_0^{3y} dn - 2 \int_0^{3y} x^2 dn - 4 \int_0^{3y} y^2 dn \right] dy$$

$$\int_0^{\sqrt{2}} \left[8 \cdot n - 2 \cdot \frac{n^3}{3} - 4y^2 \cdot (n) \right]_0^{3y} dy$$

$$\int_0^{\sqrt{2}} \left[8(3y) - \frac{2(3y)^3}{3} - 4y^2(3y) \right] dy$$

$$\int_0^{\sqrt{2}} (24y - 18y^3 - 12y^3) dy$$

$$24 \int_0^{\sqrt{2}} y dy - 18 \int_0^{\sqrt{2}} y^3 dy - 12 \int_0^{\sqrt{2}} y^3 dy$$

$$\left[24 \cdot \frac{y^2}{2} - 18 \cdot \frac{y^4}{4} - 12 \cdot \frac{y^4}{4} \right]_0^{\sqrt{2}}$$

$$\left[12 \cdot (y)^2 - \frac{9}{2} \cdot y^4 - 3 \cdot y^4 \right]_0^{\sqrt{2}}$$

$$12 \cdot (\sqrt{2})^2 - \frac{9}{2} (\sqrt{2})^4 - 3 (\sqrt{2})^4$$

$$24 - 18 - 12 = -6 \quad \text{Answer}$$

When Denominator's derivative is given in numerator, then we write that term in (ln).

09) $\int_1^e \int_1^{e^2} \int_1^{e^3} \frac{1}{xyz} dz dy dx$

Solution:-

$$\int_1^e \int_1^{e^2} \left[\frac{1}{yz} \int_1^{e^3} \frac{1}{x} dx \right] dy dz$$

$$\int_1^e \int_1^{e^2} \left[\frac{1}{yz} \left[\ln(x) \right]_1^{e^3} \right] dy dz$$

$$\int_1^e \int_1^{e^2} \left[\frac{1}{yz} \left[\ln(e^3) - \ln(1) \right] \right] dy dz$$

using logarithm rule.

$$= \int_1^e \int_1^{e^2} \left[\frac{1}{yz} [3] \right] dy dz$$

$$= 3 \int_1^e \left[\frac{1}{z} \int_1^{e^2} \frac{1}{y} dy \right] dz$$

$$3 \int_1^e \left[\frac{1}{z} \int_1^{e^2} (\ln(y)) \right] dz$$

$$3 \int_1^e \left[\frac{1}{z} (\ln(e^2) - \ln(1)) \right] dz$$

$$3 \int_1^e \left[\frac{1}{z} [2] \right] dz$$

$$3 \times 2 \int_1^e \frac{1}{z} dz$$

$$(\ln(z))_1^e$$

$$6 \left[(\ln(e) - \ln(1)) \right]$$

$$6 [1 - 0]$$

$$6(1)$$

$$= 6 \text{ Answer}$$

$$\ln e = 1$$

$$\ln(1) = 0$$

10) $\int_0^1 \int_0^{3-3n} \int_0^{3-3n-y} dz dy dn$

Solution:-

$$\int_0^1 \int_0^{3-3n} \left[\int_0^{3-3n-y} dz \right] dy dn$$

$$\int_0^1 \int_0^{3-3n} \left[z \right]_0^{3-3n-y} dy dn$$

$$\int_0^1 \int_0^{3-3n} (3-3n-y) dy dn$$

$$\int_0^1 \left[3 \int_0^{3-3n} dy - 3n \int_0^{3-3n} dy - \int_0^{3-3n} y dy \right] dn$$

$$\int_0^1 \left[3 \left[y \right]_0^{3-3n} - 3n \left[y \right]_0^{3-3n} - \left[\frac{y^2}{2} \right]_0^{3-3n} \right] dn$$

$$\int_0^1 \left[3 \cdot (3-3n) - 3n(3-3n) - \frac{(3-3n)^2}{2} \right] dn$$

$$[9-9n - 9n + 9n^2 - \frac{9}{2} + \frac{9}{2}n]$$

$$\int_0^1 \left((3-3n)^2 - \frac{(3-3n)^2}{2} \right) dn$$

$$\frac{1}{2} \int_0^1$$

$$\int_0^1 \left[(3-3n)y - \frac{1}{2} y^2 \right]_0^{3-3n} dn$$

$$(3-3n)(3-3n) - \frac{1}{2} (3-3n)^2$$

$$\frac{1}{2} [(3-3n)(3-3n) - (3-3n)^2]$$

$$\frac{1}{2} 3-3n [3-3n - (3-3n)]$$

$$\frac{1}{2} (3-3n)^2$$

$$11) \int_0^{\pi/6} \int_0^1 \int_{-2}^3 y \sin z \, dn \, dy \, dz$$

Solution:

$$= \int_0^{\pi/6} \int_0^1 \left[y \sin z \int_{-2}^3 dn \right] dy \, dz$$

$$\int_0^{\pi/6} \int_0^1 \left[y \sin z \left| n \right|_{-2}^3 \right] dy \, dz$$

$$\int_0^{\pi/6} \int_0^1 \left[y \sin z (3 - (-2)) \right] dy \, dz$$

$$5 \int_0^{\pi/6} \left[\sin z \int_0^1 y \, dy \right] dz$$

$$5 \int_0^{\pi/6} \left[\sin z \left[\frac{y^2}{2} \right]_0^1 \right] dz$$

$$5 \int_0^{\pi/6} \left(\sin z \left[\frac{1}{2} \right] \right) dz$$

$$\frac{5 \cdot 1}{2} \int_0^{\pi/6} \sin z \, dz$$

$$\frac{5}{2} \cdot \left[-\cos z \right]_0^{\pi/6}$$

$$\frac{5}{2} \cdot (\cos(\pi/6) - (-\cos(0)))$$

$$\frac{5}{2} \cdot \left(\frac{\sqrt{3}}{2} + 1 \right)$$

$$\frac{5}{4} \cdot (2 + \sqrt{3})$$

$$4 \cdot 2 - \sqrt{3}$$

$$\frac{5}{4} (2 - \sqrt{3})$$

Answer

$$12) \int_{-1}^1 \int_0^1 \int_0^2 (ny + z) \, dy \, dn \, dz$$

Solution:

$$\int_{-1}^1 \int_0^1 \left[n \int_0^2 dy + \int_0^2 y \, dy + z \int_0^2 dy \right] dn \, dz$$

$$\int_{-1}^1 \int_0^1 \left[ny + \frac{y^2}{2} + zy \right]_0^2 dn \, dz$$

$$\int_{-1}^1 \int_0^1 \left[(n(2) + \frac{(2)^2}{2} + z(2)) \right] dn \, dz$$

$$\int_{-1}^1 \int_0^1 (2n + 2 + 2z) dn \, dz$$

$$\int_{-1}^1 \left[2 \int_0^1 n \, dn + 2 \int_0^1 dn + 2z \int_0^1 dn \right] dz$$

$$\int_{-1}^1 \left[\left(\frac{2n^2}{2} + 2n + 2zn \right) \right]_0^1 dz$$

$$\int_{-1}^1 \left[n^2 + 2n + 2zn \right]_0^1 dz$$

$$(1)^2 + 2(1) + 2z(1)$$

$$1 + 2 + 2z$$

$$3 + 2z$$

$$\int_{-1}^1 (3 + 2z) dz$$

$$3 \int_{-1}^1 dz + 2 \int_{-1}^1 z \, dz$$

$$\left[3z + \frac{2z^2}{2} \right]_{-1}^1$$

$$2(1) - 2(-1) + \frac{2(1)^2}{2} - \frac{2(-1)^2}{2}$$

$$\frac{2+2}{1} + \frac{2}{2} - \frac{2}{2}$$

$$4 + 4 + 1 - 1 = 8$$

$$\left[3z + z^2 \right]_{-1}^1$$

$$(3(1) - 3(-1)) + (1)^2 - (-1)^2$$

$$3 + 3 + (1) - (1) = 6$$

$$13) \int_0^3 \int_0^{\sqrt{9-n^2}} \int_0^{\sqrt{9-n^2}} dz dy dn$$

Solution:-

$$\int_0^3 \int_0^{\sqrt{9-n^2}} \left[\int_0^{\sqrt{9-n^2}} dz \right] dy dn$$

$$\int_0^3 \int_0^{\sqrt{9-n^2}} \left[z \right]_0^{\sqrt{9-n^2}} dy dn$$

$$\int_0^3 \int_0^{\sqrt{9-n^2}} (\sqrt{9-n^2}) dy dn$$

$$\int_0^3 \left[\sqrt{9-n^2} \int_0^{\sqrt{9-n^2}} dy \right] dn$$

$$\int_0^3 \left[\sqrt{9-n^2} \cdot y \right]_0^{\sqrt{9-n^2}} dn$$

$$\int_0^3 (\sqrt{9-n^2} \cdot \sqrt{9-n^2}) dn$$

→ Radical cut

$$\int_0^3 (9-n^2) dn$$

$$9 \int_0^3 dn - \int_0^3 n^2 dn$$

$$9 \cdot n - \frac{n^3}{3}$$

$$\left[9n - \frac{n^3}{3} \right]_0^3$$

$$9(3) - \frac{(3)^3}{3}$$

$$27 - \frac{27}{3}$$

$$27 - 9$$

$$= 18$$

Answer

$$-\frac{2}{3} \left[(4-y^2)^{3/2} \right]_0^2$$

$$(4-(2)^2)^{3/2} - (4-(0)^2)^{3/2}$$

$$(4-4)^{3/2} - (4-0)^{3/2}$$

$$-\frac{2}{3}$$

$$0$$

$$-(2)^3$$

$$-8$$

$$14) \int_0^2 \int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} \int_0^{2n+y} dz dy dn$$

Solution:-

$$\int_0^2 \int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} \left[\int_0^{2n+y} dz \right] dy dn$$

$$\int_0^2 \left[z \right]_0^{2n+y} dy dn$$

$$\int_0^2 \left[2 \int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} n dy + y \int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} dy \right] dn$$

$$\int_0^2 \left[\left(2 \cdot \frac{n^2}{2} + y n \right) \right]_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} dn$$

$$\left[\frac{2n^2}{2} + ny \right]$$

upper $\int_0^2 \left[\frac{2(\sqrt{4-y^2})^2}{2} + \sqrt{4-y^2} \cdot y \right] dn$

lower $\int_0^2 \left[\frac{2(-\sqrt{4-y^2})^2}{2} + (-\sqrt{4-y^2}) \cdot y \right] dn$

$$\left(\frac{2(\sqrt{4-y^2})^2}{2} + (-\sqrt{4-y^2}) \cdot y \right)$$

Combined: (upper+lower):

$$\left(\left(\frac{2(\sqrt{4-y^2})^2}{2} + \sqrt{4-y^2} \cdot y \right) - \left(\frac{2(-\sqrt{4-y^2})^2}{2} + (-\sqrt{4-y^2}) \cdot y \right) \right)$$

$$\left((4-y^2) + (\sqrt{4-y^2})y - (4-y^2) - (-\sqrt{4-y^2})y \right)$$

$$\left(4-y^2 + \sqrt{4-y^2}y - 4+y^2 + \sqrt{4-y^2}y \right)$$

$$\left(\sqrt{4-y^2}y + \sqrt{4-y^2}y \right)$$

$$2 \sqrt{4-y^2} y$$

→ Multiple ⊖

$$+1 \sqrt{4-y^2} (-2y)$$

Derivative is given

$$-\frac{1}{1} \left(\frac{(4-y^2)^{1/2+1/1}}{\frac{1}{2}+1/1} \right) \frac{1}{3/2}$$

$$15) \int_0^1 \int_0^{2-n} \int_0^{2-n-y} dz dy dn$$

Solution:

$$\int_0^1 \int_0^{2-n} \left[\int_0^{2-n-y} dz \right] dy dn$$

$$\int_0^1 \int_0^{2-n} \left[z \right]_0^{2-n-y} dy dn$$

$$\int_0^1 \int_0^{2-n} [2-n-y] dy dn$$

$$\int_0^1 \left[\int_0^{2-n} (2-n-y) dy \right] dn$$

$$\int_0^1 \left[2y - ny - \frac{y^2}{2} \right]_0^{2-n} dn$$

$$\int_0^1 \left[2y - ny - \frac{y^2}{2} \right]_0^{2-n} dn$$

$$\int_0^1 \left[2y - ny - \frac{y^2}{2} \right]_0^{2-n} dn$$

$$\int_0^1 \left[2(2-n) - n(2-n) - \frac{(2-n)^2}{2} \right] dn$$

$$\int_0^1 \left[4 - 2n - 2n + n^2 - \frac{(2-n)^2}{2} \right] dn$$

$$\int_0^1 \left[(2-n)(2-n) - \frac{(2-n)^2}{2} \right] dn$$

$$\frac{(2-n)^2}{1} - \frac{(2-n)^2}{2}$$

$$\frac{2(2-n)^2 - (2-n)^2}{2}$$

$$\int_0^1 \frac{(2-n)^2}{2} dn$$

$$\frac{1}{2} \int_0^1 (2-n)^2 dn$$

$$\frac{1}{2} \int_0^1 z dn - \int_0^1 n dn$$

$$\left[\frac{2n - n^2}{2} \right]_0^1$$

$$\frac{2(1) - (1)^2}{2} - \frac{1}{2}$$

$$16) \int_0^1 \int_0^{1-n^2} \int_3^{4-n^2-y} n dz dy dn$$

Solution:

$$\int_0^1 \int_0^{1-n^2} \left[\int_3^{4-n^2-y} n dz \right] dy dn$$

$$\int_0^1 \int_0^{1-n^2} \left[n z \right]_3^{4-n^2-y} dy dn$$

$$\int_0^1 \int_0^{1-n^2} [n(4-n^2-y-3)] dy dn$$

$$\int_0^1 \int_0^{1-n^2} [n(1-n^2-y)] dy dn$$

$$\int_0^1 \left[(n-n^3-y n) dy \right] dn$$

$$\int_0^1 \left[n y - n^3 y - \frac{n y^2}{2} \right]_0^{1-n^2} dn$$

$$\int_0^1 \left[\left(n(1-n^2) n^3 (1-n^2) - \frac{n(1-n^2)^2}{2} \right) \right] dn$$

$$\int_0^1 \left[(1-n^2) n - n^3 - \frac{n(1-n^2)^2}{2} \right] dn$$

$$\int_0^1 \left[(1-n^2) (n - n^3) - \frac{n + n^3}{2} \right] dn$$

$$\int_0^1 \left[(1-n^2) n - n^3 - \frac{n + n^3}{2} \right] dn$$

$$\int_0^1 \left[(1-n^2) \left(\frac{n - n^3}{2} \right) \right] dn$$

$$\int_0^1 \left[(1-n^2) n \left(\frac{1-n^2}{2} \right) \right] dn$$

$$\frac{1}{2} \int_0^1 (1-n^2)(1-n^2)n$$

$$(1-n^2)^2 n$$

(uvw-space)

$$17) \int_0^\pi \int_0^\pi \int_0^\pi \cos(u+v+w) du dv dw$$

Solution:

$$\int_0^\pi \int_0^\pi \left[\int_0^\pi (\cos(u+v+w)) du \right] dv dw.$$

$$\int_0^\pi \int_0^\pi \left[\sin(u+v+w) \Big|_0^\pi \right] dv dw.$$

$$\int_0^\pi \int_0^\pi \left[(\sin(\pi+v+w) - \sin(0+v+w)) \right] dv dw$$

$$\int_0^\pi \int_0^\pi \left[(\sin(\pi+v+w) - \sin(v+w)) \right] dv dw$$

$$\int_0^\pi \left[(\sin(\pi+v+w) - \sin(v+w)) dv \right] dw.$$

$$\int_0^\pi \left[(-\cos(\pi+v+w) + \cos(v+w)) \Big|_0^\pi \right] dw$$

$$\int_0^\pi \left[\begin{aligned} & \{-\cos(\pi+\pi+w) + \cos(\pi+w)\} \\ & \{-(-\cos(\pi+0+w)) + \cos(0+w)\} \end{aligned} \right] dw$$

$$\int_0^\pi (-\cos(2\pi+w) + \cos(\pi+w) + \cos(\pi+w) + \cos(w)) dw$$

$$\int_0^\pi (-\cos(2\pi+w) + 2\cos(\pi+w) + \cos w) dw$$

$$\left(-\sin(2\pi+w) + 2\sin(\pi+w) - \sin w \right) \Big|_0^\pi$$

$$\left\{ (-\sin(2\pi+\pi) + 2\sin(\pi+\pi) - \sin \pi) \right\}$$

$$-(-\sin(2\pi+0) + 2\sin(\pi+0) - \sin 0)$$

$$-\sin(2\pi) + 2\sin(2\pi) - \sin \pi$$

$$+ \sin(2\pi) - 2\sin(\pi) + \sin 0$$

$$\boxed{=0}$$

Question #16
Last part
here

$$\frac{1}{2} \int_0^1 (1-n^2)^2 n dn$$

$$\frac{1}{2} \int_0^1 (1-2n+n^2)^2 n dn$$

$$\frac{1}{2} (1-2n^2+n^4) \cdot n$$

$$\frac{1}{2} \int_0^1 n - 2n^3 + n^5 dn$$

$$\frac{1}{2} \left[\frac{n^2}{2} - \frac{2 \cdot n^4}{4} + \frac{n^6}{6} \right]_0^1$$

$$\frac{1}{2} \left(\frac{(1)^2}{2} - \frac{(1)^4}{2} + \frac{(1)^6}{6} \right)$$

$$= \frac{1}{2} \left(\frac{1}{2} - \frac{1}{2} + \frac{1}{6} \right)$$

18) $\int_0^1 \int_1^{\sqrt{e}} \int_1^e se^s \ln r \left(\frac{\ln t}{t} \right)^2 dt dr ds$

$$-\frac{\sqrt{e}}{2} + \frac{1}{1} = \left[\frac{-\sqrt{e} + 2}{2} \right]$$

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$$\int_0^1 \int_1^{\sqrt{e}} \left[se^s \ln r \int_1^e \left(\frac{\ln t}{t} \right)^2 dt \right] dr ds$$

Integration By Parts

$uv \rightarrow \ln r$

$u = \ln r, v = 1$

$$\int_0^1 \int_1^{\sqrt{e}} \left[se^s \ln r \left(\frac{(\ln t)^{2+1}}{2+1} \right) \Big|_1^e \right] dr ds$$

$u \int v dr - \int (u' \int v dr) dr$

$\ln r \int 1 dr - \int \frac{d}{dr} \ln r \int 1 dr dr$

$$\int \int \left[se^s \ln r \left(\frac{(\ln t)^3}{3} \right) \Big|_1^e \right] dr ds$$

$$\int \int \left[se^s \ln r \left(\frac{\ln(e)^3}{3} - \frac{\ln(1)^3}{3} \right) \right] dr ds$$

$$(\ln r)r - \int \left(\frac{1}{r} \cdot r \right) dr$$

$$\int \int \left[se^s \ln r \left(\frac{1}{3} - \frac{0}{3} \right) \right] dr ds$$

$$(\ln r)r - \int 1 dr$$

$$\int \int \left[se^s \ln r \left(\frac{1}{3} \right) \right] dr ds$$

$$\boxed{(\ln r)r - r}$$

done ✓ Integration by Parts

$$\frac{1}{3} \int_0^1 \left[se^s \int_1^{\sqrt{e}} \ln r dr \right] ds$$

Again By Parts for

$se^s \Rightarrow \text{ILATE}$

$u = s, v = e^s$

$$u \int s ds - \int (u' \int s ds) ds$$

$$s \cdot e^s - \int (1 \cdot e^s) ds$$

$$se^s - \int e^s ds$$

$$\boxed{se^s - e^s}$$

done ✓ Integration by Parts

$$\frac{1}{3} \int_0^1 \left[se^s \left(r(\ln r) - r \right) \Big|_1^{\sqrt{e}} \right] ds$$

$$\frac{1}{3} \int_0^1 se^s \left(\left(\sqrt{e}(\ln \sqrt{e}) - \sqrt{e} \right) - \left(1(\ln(1) - 1) \right) \right) ds$$

$$\frac{1}{3} \int se^s \left(\frac{\sqrt{e} \cdot \ln \sqrt{e} - \sqrt{e}}{2} - (0 - 1) \right) ds$$

$$\frac{1}{3} \int se^s \left(\frac{\sqrt{e} - \sqrt{e}}{2} + 1 \right) ds$$

$$\frac{1}{3} \int se^s \left(\frac{\sqrt{e} - 2\sqrt{e}}{2} + 1 \right) ds$$

$$\frac{1}{3} \int se^s \left(\frac{-\sqrt{e} + 1}{2} \right) ds$$

$$\frac{1}{3} \left(\frac{-\sqrt{e} + 1}{2} \right) \int_0^1 se^s ds$$

$$\frac{1}{3} \left(\frac{-\sqrt{e} + 1}{2} \right) \int_0^1 se^s ds$$

By Parts

$$\left[se^s - e^s \right]_0^1$$

$$(1e^1 - e^1) - (0e^0 - e^0)$$

$$(e - e) - (0 - 1)$$

$$0 - (-1)$$

$$\frac{1}{3} \frac{-\sqrt{e} + 1}{2} \Rightarrow$$

$$\boxed{\frac{2 - \sqrt{e}}{6}}$$

Answer

$$+ 1$$

$$19) \int_0^{\pi/4} \int_0^{\sec v} \int_{-\infty}^{2t} e^n \, dn \, dt \, dv \quad (-1 \leq x) \quad 20) \int_0^7 \int_0^2 \int_0^{\sqrt{4-q^2}} \frac{q}{r+1} \, dp \, dq \, dr$$

Solution:-

$$\iint \left[\int e^n \, dn \right] dt \, dv$$

$$\iint \left[e^n \right]_{-\infty}^{2t} dt \, dv$$

$$\iint (2^{2t} - e^0) dt \, dv$$

$\downarrow$
 $0 \rightarrow \text{calculator}$

$$\iint (2^{2t}) dt \, dv$$

$$\int \left[\int e^{2t} dt \right] dv$$

$$\left[\frac{e^{2t}}{2} \right]_0^{\sec v}$$

$$\frac{1}{2} \int [e^{2 \sec v} - e^{2(0)}] dv$$

$$\frac{1}{2} \int (e^{2 \sec v} - 1) dv$$

$$\frac{1}{2} \int (\sec^2 v - 1) dv$$



$\downarrow$ integration?
 $\sec^2 v = \tan v$
 $1 = v$

$$\frac{1}{2} \int (\tan v - v) \Big|_0^{\pi/4}$$

$$\frac{1}{2} [(\tan \pi/4 - \pi/4) - (\tan(0) - 0)]$$

$$\frac{1}{2} [(1 - \pi/4) - (0 - 0)]$$

$$\frac{1}{2} (1 - \pi/4)$$

$$\boxed{\frac{1}{2} - \frac{\pi}{8}}$$

Answer

Solution:-

$$\iint \left[\frac{q}{r+1} \right] dp \, dq \, dr$$

$$\iint \left[\frac{q}{r+1} p \right]_0^{\sqrt{4-q^2}} dq \, dr$$

$$\iint \left[\frac{q}{r+1} \sqrt{4-q^2} - 0 \right] dq \, dr$$